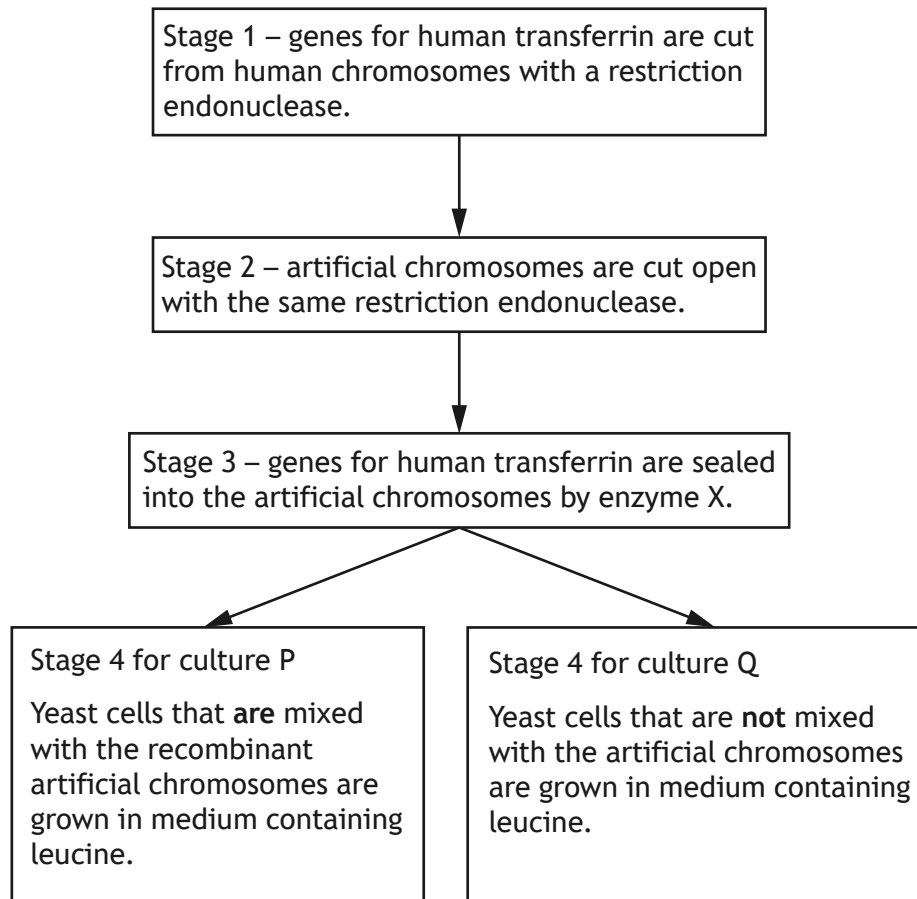


10. Transferrin is a protein used to treat the blood disorder anaemia in humans.

Artificial chromosomes are used in recombinant DNA technology to genetically modify yeast cells to produce transferrin.

The strain of yeast cells used cannot synthesise the amino acid leucine, which is necessary for protein synthesis and growth of the yeast.

Some stages of this recombinant DNA technology process are shown in the diagram.



10. (continued)

- (a) (i) State the term used to describe the artificial chromosome that carries the human gene into yeast cells.

1

- (ii) State why an artificial chromosome rather than a plasmid may be used in recombinant DNA technology.

1

- (b) Explain why the same restriction endonuclease is used in stages 1 and 2.

1

- (c) Name enzyme X used in stage 3.

1

- (d) The artificial chromosome used contains a selectable marker gene that only allows transformed yeast cells to synthesise leucine.

After stage 4, cultures P and Q were transferred to separate plates containing solid medium without leucine.

After incubation, yeast cells only grew on the plate containing culture P.

Explain this result.

2

[Turn over

